

**SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING**
**Product Identifier**

**Product Description:** Hexafluoro-2-butyne  
**Cat No. :** S60353  
**CAS No** 692-50-2  
**Molecular Formula** C4F6

**Relevant identified uses of the substance or mixture and uses advised against**

**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

**Details of the supplier of the safety data sheet**
**Importer**

Fisher Scientific Korea  
 D5,D6, Incheon Airport Logistics Complex  
 150, Gonghangdong-Ro 296 Beon-Gil  
 Jung-Gu, Incheon  
 Tel: +82-1661-9555  
 Fax: +82-2-2023-0603

**Supplier**

Thermo Fisher Scientific Chemicals, Inc.  
 30 Bond Street  
 Ward Hill, MA 01835-8099

**E-mail address** Chem.KR@thermofisher.com

**Emergency Telephone Number**

Emergency telephone: Medical: +(82) 070-7686-0086 or + 1-703-741-5970  
 CHEMTREC: 080 822 1374 (Local), CHEMTREC: 1-800-424-9300 or + 1-703-527-3887  
 Korea: 00-308-13-2549 (24 hours a day, 7 days a week)

**SECTION 2: HAZARDS IDENTIFICATION**
**Classification of the substance or mixture**
**Physical hazards**

Flammable gases  
 Gases under pressure

Category 2  
 Liquefied gas

**Health hazards**

Acute Inhalation Toxicity - Gas

Category 3

**Environmental hazards**

Based on available data, the classification criteria are not met

**Label Elements**

# SAFETY DATA SHEET

Hexafluoro-2-butyne

Revision Date 26-Feb-2024



Signal Word

Danger

## Hazard Statements

H221 - Flammable gas

H280 - Contains gas under pressure; may explode if heated

H331 - Toxic if inhaled

## Precautionary Statements

### Prevention

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

P261 - Avoid breathing dust/fume/gas/mist/vapors/spray

P271 - Use only outdoors or in a well-ventilated area

### Response

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing

P311 - Call a POISON CENTER or doctor

P377 - Leaking gas fire: Do not extinguish, unless leak can be stopped safely

P381 - In case of leakage, eliminate all ignition sources

### Storage

P403 - Store in a well-ventilated place

### Disposal

P502 - Refer to manufacturer or supplier for information on recovery or recycling

## Other Hazards

This product does not contain any known or suspected endocrine disruptors

## NFPA

Health  
3

Flammability  
3

Instability  
0

Physical hazards  
-

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

### 3.1. Substances

| Component                         | Common Name              | CAS No   | Index No   | Weight % |
|-----------------------------------|--------------------------|----------|------------|----------|
| 2-Butyne, 1,1,1,4,4,4-hexafluoro- | No information available | 692-50-2 | Not listed | 99 - 100 |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

#### General Advice

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

#### Eye Contact

In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.

#### Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. Immediate medical

# SAFETY DATA SHEET

Hexafluoro-2-butyne

Revision Date 26-Feb-2024

attention is required.

## Ingestion

Do NOT induce vomiting. Call a physician or poison control center immediately.

## Inhalation

If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Remove to fresh air. Immediate medical attention is required.

## Self-Protection of the First Aider

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

## Most important symptoms and effects, both acute and delayed

Difficulty in breathing. Symptoms of overexposure are dizziness, headache, tiredness, nausea, unconsciousness, cessation of breathing.

## Indication of any immediate medical attention and special treatment needed

### Notes to Physician

Treat symptomatically.

## SECTION 5: FIREFIGHTING MEASURES

### Extinguishing media

#### Suitable Extinguishing Media

CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam.

#### Extinguishing media which must not be used for safety reasons

No information available.

### Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors. Risk of explosion if heated under confinement.

### Hazardous Combustion Products

Carbon oxides, Hydrogen fluoride.

### Advice for fire-fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### Personal Precautions, Protective Equipment and Emergency Procedures

Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak.

### Environmental precautions

Prevent further leakage or spillage if safe to do so.

### Methods and Material for Containment and Cleaning Up

Ventilate the area. Prevent further leakage or spillage if safe to do so. Wear self-contained breathing apparatus and protective suit. Remove all sources of ignition.

### Reference to Other Sections

# SAFETY DATA SHEET

Hexafluoro-2-butyne

Revision Date 26-Feb-2024

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

Use only under a chemical fume hood. Wear personal protective equipment/face protection. Do not breathe gas. Do not get in eyes, on skin, or on clothing.

### Conditions for Safe Storage, Including any Incompatibilities

Corrosives area. Keep containers tightly closed in a dry, cool and well-ventilated place.

### Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

| Component                            | CAS No   | Korea      | ACGIH TLV  | OSHA PEL   |
|--------------------------------------|----------|------------|------------|------------|
| 2-Butyne,<br>1,1,1,4,4,4-hexafluoro- | 692-50-2 | Not listed | Not listed | Not listed |

| Component                            | CAS No   | European Union | The United Kingdom | Germany    |
|--------------------------------------|----------|----------------|--------------------|------------|
| 2-Butyne,<br>1,1,1,4,4,4-hexafluoro- | 692-50-2 | Not listed     | Not listed         | Not listed |

### ACGIH - Biological Exposure Indices

| Component                            | CAS No   | ACGIH - Biological Exposure Indices |
|--------------------------------------|----------|-------------------------------------|
| 2-Butyne,<br>1,1,1,4,4,4-hexafluoro- | 692-50-2 | Not listed                          |

### Exposure Controls

#### Engineering Measures

Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

#### Personal protective equipment

##### Eye Protection

Goggles

##### Hand Protection

Protective gloves

##### Skin and body protection

Long sleeved clothing

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.

(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

##### Personal protective equipment

##### Respiratory Protection

Use only those certified by the Korea Occupational Safety and Health Administration.

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators

##### Recommended Filter type:

Organic gases and vapours filter

# SAFETY DATA SHEET

Hexafluoro-2-butyne

Revision Date 26-Feb-2024

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly  
When RPE is used a face piece Fit Test should be conducted

## Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice

## Environmental exposure controls

No information available

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

Appearance (Physical State, Color, etc.) Colorless Liquid

Odor pungent

Odor Threshold No data available

pH No information available

Melting Point/Range -117 °C / -178.6 °F

Literature reference

Softening Point No data available

Boiling Point/Range -25 °C / -13 °F

Literature reference

Flash Point No data available

Method - No information available

Evaporation Rate No data available

Flammability (solid,gas) Not applicable

Liquid

Explosion Limits No data available

Vapor Pressure 7,237.4 hPa @ 20°C

Vapor Density No data available

(Air = 1.0)

Specific Gravity / Density No data available

Bulk Density Not applicable

Liquid

Water Solubility Insoluble

Solubility in other solvents No information available

### Partition Coefficient (n-octanol/water)

| Component                         | CAS No   | log Pow           |
|-----------------------------------|----------|-------------------|
| 2-Butyne, 1,1,1,4,4,4-hexafluoro- | 692-50-2 | No data available |

Autoignition Temperature No data available

Decomposition Temperature No data available

Viscosity No data available

Explosive Properties No information available

Oxidizing Properties No information available

Molecular Formula C4F6

Molecular Weight 162.03

## SECTION 10: STABILITY AND REACTIVITY

# SAFETY DATA SHEET

Hexafluoro-2-butyne

Revision Date 26-Feb-2024

## Reactivity

None known, based on information available

## Chemical Stability

Stable under normal conditions.

## Possibility of Hazardous Reactions

### **Hazardous Polymerization**

No information available.

### **Hazardous Reactions**

None under normal processing.

## Conditions to Avoid

None known.

## Incompatible Materials

None known.

## Hazardous Decomposition Products

Carbon oxides. Hydrogen fluoride.

## SECTION 11: TOXICOLOGICAL INFORMATION

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### Product Information

#### Information on expected route of exposure

##### **Inhalation**

Toxic by inhalation.

##### **Ingestion**

May be harmful if swallowed.

##### **Eyes**

Avoid contact with eyes.

##### **Skin**

Avoid contact with skin.

#### Information on Health Hazards

##### **(a) acute toxicity;**

###### **Oral**

No data available

###### **Dermal**

No data available

###### **Inhalation**

Category 3

| Component                         | CAS No   | LD50 Oral         | LD50 Dermal       | LC50 Inhalation              |
|-----------------------------------|----------|-------------------|-------------------|------------------------------|
| 2-Butyne, 1,1,1,4,4,4-hexafluoro- | 692-50-2 | No data available | No data available | LC50 = 81 ppm ( Rat )<br>4 h |

**(b) skin corrosion/irritation;** No data available

**(c) serious eye damage/irritation;** No data available

##### **(d) respiratory or skin sensitization;**

###### **Respiratory**

No data available

###### **Skin**

No data available

| Component                         | CAS No   | Test method       | Test species      | Study result      |
|-----------------------------------|----------|-------------------|-------------------|-------------------|
| 2-Butyne, 1,1,1,4,4,4-hexafluoro- | 692-50-2 | No data available | No data available | No data available |

**(e) germ cell mutagenicity;** No data available

# SAFETY DATA SHEET

Hexafluoro-2-butyne

Revision Date 26-Feb-2024

| Component                         | CAS No   | Test method       | Test species      | Study result      |
|-----------------------------------|----------|-------------------|-------------------|-------------------|
| 2-Butyne, 1,1,1,4,4,4-hexafluoro- | 692-50-2 | No data available | No data available | No data available |

(f) carcinogenicity; No data available

| Component                         | CAS No   | Test method       | Test species / Duration | Study result      |
|-----------------------------------|----------|-------------------|-------------------------|-------------------|
| 2-Butyne, 1,1,1,4,4,4-hexafluoro- | 692-50-2 | No data available | No data available       | No data available |

There are no known carcinogenic chemicals in this product

| Component                         | CAS No   | IARC       | NTP        | ACGIH      | OSHA       | UK         |
|-----------------------------------|----------|------------|------------|------------|------------|------------|
| 2-Butyne, 1,1,1,4,4,4-hexafluoro- | 692-50-2 | Not listed | Not listed | Not listed | Not listed | Not listed |

(g) reproductive toxicity; No data available

| Component                         | CAS No   | Test method       | Test species / Duration | Study result      |
|-----------------------------------|----------|-------------------|-------------------------|-------------------|
| 2-Butyne, 1,1,1,4,4,4-hexafluoro- | 692-50-2 | No data available | No data available       | No data available |

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available

Target Organs None known.

(j) aspiration hazard; No data available

## Other Adverse Effects

Symptoms of overexposure are dizziness, headache, tiredness, nausea, unconsciousness, cessation of breathing.

| Component                         | CAS No   | EU - Endocrine Disruptors Candidate List | EU - Endocrine Disruptors - Evaluated Substances | Japan - Endocrine Disruptor Information |
|-----------------------------------|----------|------------------------------------------|--------------------------------------------------|-----------------------------------------|
| 2-Butyne, 1,1,1,4,4,4-hexafluoro- | 692-50-2 | Not applicable                           | Not applicable                                   | Not applicable                          |

## SECTION 12: ECOLOGICAL INFORMATION

**Ecotoxicity effects** Contains no substances known to be hazardous to the environment or that are not degradable in waste water treatment plants.

| Component                         | CAS No   | Freshwater Fish   | Water Flea        | Freshwater Algae  | Microtox          |
|-----------------------------------|----------|-------------------|-------------------|-------------------|-------------------|
| 2-Butyne, 1,1,1,4,4,4-hexafluoro- | 692-50-2 | No data available | No data available | No data available | No data available |

## Persistence and degradability

**Persistence** May persist, based on information available.

**Bioaccumulative potential** Product has a high potential to bioconcentrate

**Mobility in soil** The product contains volatile organic compounds (VOC) which will evaporate easily from all surfaces. Will likely be mobile in the environment due to its volatility. Disperses rapidly in air.

## Ozone Depletion Potential

ALFAAS60353

# SAFETY DATA SHEET

Hexafluoro-2-butyne

Revision Date 26-Feb-2024

| Component                         | CAS No   | Ozone Depletion Potential |
|-----------------------------------|----------|---------------------------|
| 2-Butyne, 1,1,1,4,4,4-hexafluoro- | 692-50-2 | Not listed                |

Other adverse effects No information available

## SECTION 13: DISPOSAL CONSIDERATIONS

### Waste treatment methods

|                                            |                                                                                                                                                                                                                                                                                                                                            |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste from Residues/Unused Products</b> | Waste is classified as hazardous. Dispose in accordance with the Wastes Control Act (폐기물관리법).                                                                                                                                                                                                                                              |
| <b>Contaminated Packaging</b>              | Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.                                                                                                   |
| <b>Other Information</b>                   | Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Can be landfilled or incinerated, when in compliance with local regulations. Do not empty into drains. Large amounts will affect pH and harm aquatic organisms. Do not let this chemical enter the environment. |

## SECTION 14: TRANSPORT INFORMATION

### Road and Rail Transport

|                         |                                         |
|-------------------------|-----------------------------------------|
| UN-No                   | UN3160                                  |
| Proper Shipping Name    | LIQUEFIED GAS, TOXIC, FLAMMABLE, N.O.S. |
| Technical Shipping Name | 1,1,1,4,4,4-Hexafluorobut-2-yne         |
| Hazard Class            | 2                                       |
| Subsidiary Hazard Class | 2.1                                     |

### IATA

|                         |                                                                 |
|-------------------------|-----------------------------------------------------------------|
| Technical Shipping Name | FORBIDDEN FOR IATA TRANSPORT<br>1,1,1,4,4,4-Hexafluorobut-2-yne |
|-------------------------|-----------------------------------------------------------------|

### IMDG/IMO

|                         |                                                                                                          |
|-------------------------|----------------------------------------------------------------------------------------------------------|
| UN-No                   | UN3160                                                                                                   |
| Proper Shipping Name    | LIQUEFIED GAS, TOXIC, FLAMMABLE, N.O.S.                                                                  |
| Technical Shipping Name | 1,1,1,4,4,4-Hexafluorobut-2-yne                                                                          |
| Hazard Class            | 2                                                                                                        |
| Subsidiary Hazard Class | 2.1                                                                                                      |
| Marine Pollutant        | Dangerous for the environment<br>Product is a marine pollutant according to the criteria set by IMDG/IMO |

**Special Precautions for User** No special precautions required

## SECTION 15: REGULATORY INFORMATION

### Safety, health and environmental regulations/legislation specific for the substance or mixture

**Legend:** X - Listed '-' - Not Listed

### International Inventories

| Component                         | CAS No   | KECL | TSCA | EINECS    | IECSC | DSL | NDSL | PICCS | ENCS | ISHL | AICS |
|-----------------------------------|----------|------|------|-----------|-------|-----|------|-------|------|------|------|
| 2-Butyne, 1,1,1,4,4,4-hexafluoro- | 692-50-2 | -    | X    | 211-732-7 | -     | -   | X    | -     | -    |      | -    |

| Component | CAS No | Seveso III Directive | Seveso III Directive | Rotterdam | Basel Convention |
|-----------|--------|----------------------|----------------------|-----------|------------------|
|-----------|--------|----------------------|----------------------|-----------|------------------|



# SAFETY DATA SHEET

Hexafluoro-2-butyne

Revision Date 26-Feb-2024

|                                      |          | (2012/18/EC) -<br>Qualifying Quantities<br>for Major Accident<br>Notification | (2012/18/EC) -<br>Qualifying Quantities<br>for Safety Report<br>Requirements | Convention (PIC) | (Hazardous Waste) |
|--------------------------------------|----------|-------------------------------------------------------------------------------|------------------------------------------------------------------------------|------------------|-------------------|
| 2-Butyne,<br>1,1,1,4,4,4-hexafluoro- | 692-50-2 | Not applicable                                                                | Not applicable                                                               | Not applicable   | Not applicable    |

| Component                         | CAS No   | OECD HPV       | Persistent Organic<br>Pollutant | Ozone Depletion<br>Potential |
|-----------------------------------|----------|----------------|---------------------------------|------------------------------|
| 2-Butyne, 1,1,1,4,4,4-hexafluoro- | 692-50-2 | Not applicable | Not applicable                  | Not applicable               |

## Korean National Regulations

| Component                         | CAS No   | Act on Registration and<br>Evaluation of Chemical<br>Substances (K-REACH) | Authorised Chemicals | Existing Substances<br>Subject to Registration |
|-----------------------------------|----------|---------------------------------------------------------------------------|----------------------|------------------------------------------------|
| 2-Butyne, 1,1,1,4,4,4-hexafluoro- | 692-50-2 | Not applicable                                                            | Not applicable       | Not applicable                                 |

| Component                         | CAS No   | Chemical Control Act -<br>Toxic Chemicals | Chemical Control Act -<br>Prohibited Chemicals | Chemical Control Act -<br>Use Restricted<br>Chemicals |
|-----------------------------------|----------|-------------------------------------------|------------------------------------------------|-------------------------------------------------------|
| 2-Butyne, 1,1,1,4,4,4-hexafluoro- | 692-50-2 | Not applicable                            | Not applicable                                 | Not applicable                                        |

| Component                         | CAS No   | Chemical Control Act -<br>Accident Precaution<br>Chemicals (% in<br>mixtures) | Chemical Control Act -<br>Accident Precaution<br>Chemicals - Quantity<br>limits Storage (% in<br>mixtures) | Chemical Control Act -<br>Accident Precaution<br>Chemicals - Quantity<br>limits Manufacture/Use<br>(% in mixtures) |
|-----------------------------------|----------|-------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| 2-Butyne, 1,1,1,4,4,4-hexafluoro- | 692-50-2 | Not applicable                                                                | Not applicable                                                                                             | Not applicable                                                                                                     |

| Component                         | CAS No   | Waste Control Law | Ministry of Environment -<br>CMR risk | Ministry of Environment -<br>Critically Controlled<br>Substance |
|-----------------------------------|----------|-------------------|---------------------------------------|-----------------------------------------------------------------|
| 2-Butyne, 1,1,1,4,4,4-hexafluoro- | 692-50-2 | Not applicable    | Not applicable                        | Not applicable                                                  |

CCA = Chemical Control Act

ISHA = Subject to Process Safety Reports

| Component                         | CAS No   | ISHA - Harmful Agents<br>Subject to Work<br>Environment Monitoring | ISHA - Prohibited<br>substances | ISHA - Substances<br>requiring permission |
|-----------------------------------|----------|--------------------------------------------------------------------|---------------------------------|-------------------------------------------|
| 2-Butyne, 1,1,1,4,4,4-hexafluoro- | 692-50-2 | Not applicable                                                     | Not applicable                  | Not applicable                            |

| Component                         | CAS No   | ISHA - Substances<br>subject to control | ISHA - Harmful Agents<br>Requiring Health<br>Examination | ISHA - Permissible<br>Exposure Limits |
|-----------------------------------|----------|-----------------------------------------|----------------------------------------------------------|---------------------------------------|
| 2-Butyne, 1,1,1,4,4,4-hexafluoro- | 692-50-2 | Not applicable                          | Not applicable                                           | Not applicable                        |

| Component                         | CAS No   | ISHA - Subject to<br>Process Safety Reports<br>(minimum quantity) | ISHA - Threshold Limit<br>Values (TLVs) Chemicals | ISHA - Special<br>management materials |
|-----------------------------------|----------|-------------------------------------------------------------------|---------------------------------------------------|----------------------------------------|
| 2-Butyne, 1,1,1,4,4,4-hexafluoro- | 692-50-2 | Not applicable                                                    | Not applicable                                    | Not applicable                         |

## National Fire Association - Dangerous Substances Minimum quantity requiring a permit

| Component | CAS No   | Class 1 -<br>Oxidising<br>solids | Class 2 -<br>Flammable<br>solid | Class 3 -<br>Spontaneously<br>Combustible<br>Substances<br>and Dangerous<br>Substances<br>When Wet | Class 4 -<br>Flammable<br>liquids | Class 5 -<br>Self-reactive<br>substances | Class 6 -<br>Oxidising<br>liquids |
|-----------|----------|----------------------------------|---------------------------------|----------------------------------------------------------------------------------------------------|-----------------------------------|------------------------------------------|-----------------------------------|
| 2-Butyne, | 692-50-2 | Not applicable                   | Not applicable                  | Not applicable                                                                                     | Not applicable                    | Not applicable                           | Not applicable                    |

# SAFETY DATA SHEET

Hexafluoro-2-butyne

Revision Date 26-Feb-2024

|                         |  |  |  |  |  |  |  |
|-------------------------|--|--|--|--|--|--|--|
| 1,1,1,4,4,4-hexafluoro- |  |  |  |  |  |  |  |
|-------------------------|--|--|--|--|--|--|--|

## NFA Dangerous Substances Classification

Class 4.5 (Group 3 petroleum, soluble) - 4000 L

## Control Parameters

| Component                         | CAS No   | Korea      | ACGIH - Biological Exposure Indices |
|-----------------------------------|----------|------------|-------------------------------------|
| 2-Butyne, 1,1,1,4,4,4-hexafluoro- | 692-50-2 | Not listed | Not listed                          |

## US Management Information

OSHA - Occupational Safety and Health Administration

Not applicable

| Component                         | CAS No   | Specifically Regulated Chemicals | Highly Hazardous Chemicals |
|-----------------------------------|----------|----------------------------------|----------------------------|
| 2-Butyne, 1,1,1,4,4,4-hexafluoro- | 692-50-2 | Not applicable                   | Not applicable             |

## CERCLA

This material, as supplied, contains one or more substances regulated as a hazardous substance under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355)

| Component                         | CAS No   | CERCLA Extremely Hazardous Substances RQs | Hazardous Substances RQs | SARA 313 - Threshold Values % |
|-----------------------------------|----------|-------------------------------------------|--------------------------|-------------------------------|
| 2-Butyne, 1,1,1,4,4,4-hexafluoro- | 692-50-2 | Not applicable                            | Not applicable           | Not applicable                |

## CLP Classification - According to GB-CLP Regulations UK SI 2019/720 and UK SI 2020/1567

Danger.

H221 - Flammable gas. H280 - Contains gas under pressure; may explode if heated. H331 - Toxic if inhaled.

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking. P280 - Wear protective gloves/protective clothing/eye protection/face protection. P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing. P311 - Call a POISON CENTER or doctor/physician. P377 - Leaking gas fire: Do not extinguish, unless leak can be stopped safely. P381 - In case of leakage, eliminate all ignition sources. P403 - Store in a well-ventilated place.

## SECTION 16: OTHER INFORMATION

### Legend

CAS - Chemical Abstracts Service

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

NZIoC - New Zealand Inventory of Chemicals

WEL - Workplace Exposure Limit

ACGIH - American Conference of Governmental Industrial Hygienists

RPE - Respiratory Protective Equipment

LC50 - Lethal Concentration 50%

POW - Partition coefficient Octanol:Water

TWA - Time Weighted Average

IARC - International Agency for Research on Cancer

LD50 - Lethal Dose 50%

EC50 - Effective Concentration 50%

ADR - European Agreement Concerning the International Carriage of Dangerous Goods by Road

IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code

OECD - Organisation for Economic Co-operation and Development

ICAO/IATA - International Civil Aviation Organization/International Air Transport Association

MARPOL - International Convention for the Prevention of Pollution from Ships

ATE - Acute Toxicity Estimate

# SAFETY DATA SHEET

Hexafluoro-2-butyne

Revision Date 26-Feb-2024

BCF - Bioconcentration factor

VOC - (Volatile Organic Compound)

## Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

## Training Advice

Chemical incident response training.

|                  |                                             |
|------------------|---------------------------------------------|
| Prepared By      | Health, Safety and Environmental Department |
| Creation Date    | 26-Feb-2024                                 |
| Revision Date    | 26-Feb-2024                                 |
| Revision Number  | 1                                           |
| Revision Summary | Initial Release.                            |

## MOEL's Public Notice No. 2023-9 (Standards for Classification and Labeling of Chemical Substances and Safety Data Sheets)

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**